

STAT204 - HW01

Setup

Loading Relevant Package

```
library(palmerpenguins)
```

Attaching package: 'palmerpenguins'

The following objects are masked from 'package:datasets':

penguins, penguins_raw

Exploring “penguin” Data File

```
summary(penguins)
```

species	island	bill_length_mm	bill_depth_mm
Adelie :152	Biscoe :168	Min. :32.10	Min. :13.10
Chinstrap: 68	Dream :124	1st Qu.:39.23	1st Qu.:15.60
Gentoo :124	Torgersen: 52	Median :44.45	Median :17.30
		Mean :43.92	Mean :17.15
		3rd Qu.:48.50	3rd Qu.:18.70
		Max. :59.60	Max. :21.50
		NA's :2	NA's :2
flipper_length_mm	body_mass_g	sex	year
Min. :172.0	Min. :2700	female:165	Min. :2007

1st Qu.:190.0	1st Qu.:3550	male :168	1st Qu.:2007
Median :197.0	Median :4050	NA's : 11	Median :2008
Mean :200.9	Mean :4202		Mean :2008
3rd Qu.:213.0	3rd Qu.:4750		3rd Qu.:2009
Max. :231.0	Max. :6300		Max. :2009
NA's :2	NA's :2		

```
## to check how many rows are in the dataframe
nrow(penguins)
```

```
[1] 344
```

```
## to check how many columns are in the dataframe
ncol(penguins)
```

```
[1] 8
```

```
## to check what variable types are in the dataframe
str(penguins)
```

```
tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
 $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g    : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex           : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
 $ year          : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

Part 1: Vectors and Summary Statistics

1.1 Extract and Analyze Bill Lengths

```
## Extract bill_length_mm as a vector
## this pulls out the data in the "bill_length_mm" column
bill_lengths <- penguins$bill_length_mm
```

```
## Calculate: mean, median, sd, min, max (handle NAs!)
mean_bill_length <- mean(bill_lengths, na.rm = TRUE)
median_bill_length <- median(bill_lengths, na.rm = TRUE)
sd_bill_length <- sd(bill_lengths, na.rm = TRUE)
min_bill_length <- min(bill_lengths, na.rm = TRUE)
max_bill_length <- max(bill_lengths, na.rm = TRUE)
```

1.2 Logical Subsetting

```
# Create logical vector: TRUE when bill length > 45mm
long_bills <- penguins$bill_length_mm > 45

# How many penguins have long bills?
sum(long_bills, na.rm = TRUE) ## 165 penguins have bills longer than 45mm
```

```
[1] 165
```

Part 2: Subsetting Data

2.1 Island analysis

```
## to check how many different island names are in the column
unique(penguins$island)
```

```
[1] Torgersen Biscoe Dream
Levels: Biscoe Dream Torgersen
```

```
# Extract bill lengths for Dream island penguins only
dream_bill_lengths <- penguins$bill_length_mm[penguins$island == "Dream"]
dream_bill_lengths
```

```
[1] 39.5 37.2 39.5 40.9 36.4 39.2 38.8 42.2 37.6 39.8 36.5 40.8 36.0 44.1 37.0
[16] 39.6 41.1 37.5 36.0 42.3 37.3 41.3 36.3 36.9 38.3 38.9 35.7 41.1 34.0 39.6
[31] 36.2 40.8 38.1 40.3 33.1 43.2 36.8 37.5 38.1 41.1 35.6 40.2 37.0 39.7 40.2
[46] 40.6 32.1 40.7 37.3 39.0 39.2 36.6 36.0 37.8 36.0 41.5 46.5 50.0 51.3 45.4
[61] 52.7 45.2 46.1 51.3 46.0 51.3 46.6 51.7 47.0 52.0 45.9 50.5 50.3 58.0 46.4
```

```
[76] 49.2 42.4 48.5 43.2 50.6 46.7 52.0 50.5 49.5 46.4 52.8 40.9 54.2 42.5 51.0
[91] 49.7 47.5 47.6 52.0 46.9 53.5 49.0 46.2 50.9 45.5 50.9 50.8 50.1 49.0 51.5
[106] 49.8 48.1 51.4 45.7 50.7 42.5 52.2 45.2 49.3 50.2 45.6 51.9 46.8 45.7 55.8
[121] 43.5 49.6 50.8 50.2
```

```
# Calculate the mean for this subset
mean(dream_bill_lengths, na.rm = TRUE)
```

```
[1] 44.16774
```

2.2 Species Counting

```
# How many Adelie penguins are in the dataset?
adelie_count <- sum(penguins$species == "Adelie")
adelie_count
```

```
[1] 152
```

2.3 Find the Heaviest Penguin

```
# Find: body mass, species, and island of the heaviest penguin

## to identify row index (or location) of the maximum body mass in the dataframe
big_penguin_row <- which.max(penguins$body_mass_g)
big_penguin_row
```

```
[1] 170
```

```
## extract information from specific columns in the row
penguins[big_penguin_row, c("body_mass_g", "species", "island")]
```

```
# A tibble: 1 x 3
  body_mass_g species island
    <int> <fct>   <fct>
1     6300 Gentoo  Biscoe
```

Part 3: Matrices and apply()

3.1 Create Measurement Matrix

```
# Remove rows with missing values
penguins_complete <- penguins[complete.cases(penguins), ]

# Create matrix with: bill_length_mm, bill_depth_mm,
# flipper_length_mm, body_mass_g

## column bind function to combine vectors side by side as columns in a matrix
penguin_measurements <- cbind(
  penguins$bill_length_mm,
  penguins$bill_depth_mm,
  penguins$flipper_length_mm,
  penguins$body_mass_g
)

## column names assigns names to columns in new matrix
colnames(penguin_measurements) <- c(
  "bill_length_mm",
  "bill_depth_mm",
  "flipper_length_mm",
  "body_mass_g"
)
```

3.2 Apply Functions

```
# Use apply() to calculate mean of each column
apply(penguin_measurements, 2, mean, na.rm = TRUE) ## the number 2 is for columns
```

bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
43.92193	17.15117	200.91520	4201.75439

```
# Use apply() to find which row has the maximum for each column
## which.max finds position of max value
## note this is not the actual max value in each column
apply(penguin_measurements, 2, which.max)
```

bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
186	20	216	170

Part 4: Custom Function

```
# Function that: removes NAs, returns (x - mean) / sd
## this creates a standardizing function called "standardize"

standardize <- function(x) {
  x <- na.omit(x)          # this removes NAs
  (x - mean(x)) / sd(x)    # this standardizes each value
}

# Test on bill_length_mm
bill_length_standardized <- standardize(penguins$bill_length_mm)

# Verify: mean 0, sd 1
mean(bill_length_standardized)
```

```
[1] -6.882084e-17
```

```
sd(bill_length_standardized)
```

```
[1] 1
```

Bonus

Option A: Create a summary data frame with one row per species showing: count, mean body mass, mean flipper length

```
## first need to create 4 vectors

## 1. to get the unique number of species
species <- unique(penguins$species)

## 2. to count the number of penguins in each species
penguin_count <- table(penguins$species)
```

```

## 3. to calculate mean body mass for each species
mean_body_mass <- tapply(
  penguins$body_mass_g,    ## to specify the variable
  penguins$species,        ## to specify how it should be grouped (e.g. by species)
  mean,                   ## to specify desired function
  na.rm = TRUE            ## to remove NAs
)

## 4. to calculate mean flipper length for each species
mean_flipper_length <- tapply(
  penguins$flipper_length_mm, ## to specify the variable
  penguins$species,          ## to specify how it should be grouped
  mean,                     ## to specify desired function
  na.rm = TRUE              ## to remove NAs
)

## summary dataframe
penguins_summary_df <- data.frame(
  species = species,        ## left side is the name of column
  count = penguin_count,    ## right side is the object containing the data
  mean_body_mass = mean_body_mass,
  mean_flipper_length = mean_flipper_length
)

```